Assignment No: 8

Visualization of Real-Time Application Metrics Using Grafana

Aim

To configure Grafana dashboards for visualizing real-time application metrics collected by Prometheus.

Output

1. Verify cluster

```
aditya@aditya:~/devops-task-manager$ minikube start
* Done! kubectl is now configured to use "minikube" cluster
aditya@aditya:~/devops-task-manager$ kubectl get pods
task-manager-deployment-yyyyy-aaaaa   1/1   Running     0 1h
task-manager-deployment-yyyyy-bbbbb   1/1   Running     0 1h
prometheus-deployment-zzzzz-aaaaa     1/1   Running     0 1h
aditya@aditya:~/devops-task-manager$ kubectl get svc
NAME                   TYPE       PORT(S)           AGE
task-manager-service   NodePort   3000:31245/TCP    1h
prometheus-service     NodePort   9090:30090/TCP    1h
```

2. Deploy Grafana

```
aditya@aditya:~/devops-task-manager$ kubectl apply -f grafana-deployment.yaml
deployment.apps/grafana-deployment created
aditya@aditya:~/devops-task-manager$ kubectl apply -f grafana-service.yaml
service/grafana-service created
aditya@aditya:~/devops-task-manager$ kubectl get pods
grafana-deployment-wwwww-aaaaa   1/1   Running   0   40s
aditya@aditya:~/devops-task-manager$ minikube service grafana-service
* Opening service default/grafana-service in default browser...
```

3. Data source

```
Connections > Data Sources > Add data source > Prometheus
URL: http://prometheus-service:9090
Save & Test
Successfully queried the Prometheus API.
```

4. Dashboard panels

```
Dashboard: Student Task Manager Monitoring Dashboard
Application Status    : up{job="task-manager"}                                (Stat)
HTTP Request Rate     : sum(rate(task_manager_http_requests_total[1m]))       (Time series)
Process Memory Usage : process_resident_memory_bytes                          (Time series, Bytes)
Node.js Heap Memory   : nodejs_heap_size_used_bytes                           (Gauge)
Refresh interval      : 5s
```

5. Dashboard output

```
Student Task Manager Monitoring Dashboard

Application Status
UP

HTTP Request Rate
0.35 requests/second

Process Memory Usage
65 MB

Node.js Heap Memory
8.2 MB

Dashboard Refresh
5 Seconds

Grafana Dashboard Successfully Configured
The dashboard displays real-time application metrics collected by Prometheus.
```

Conclusion

Grafana was connected to Prometheus and a dashboard was built to visualize application status, request rate and memory usage in real time.